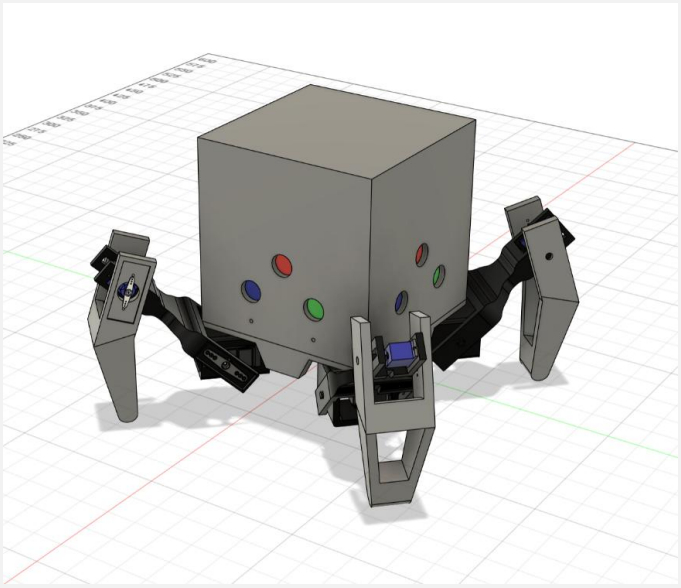


QUADRUPED **ROBOT PROJECT**

GAME INSPIRED DESIGN

Abstract

The objective of this project is to create a four-legged robot inspired by an enemy design from the game "Tunic." I plan to use a Raspberry Pi with Python script to program and control the legs, each of which is using three servo motors to provide three degrees of freedom per leg. Though the purpose of this project is to better my design and programming skills, I will consider the design to be a success when the robot can walk forwards, backwards, to the sides, and reorient itself. *(A Note for the Reader: This project is actively ongoing. Not all information or design decisions are final, but this currently serves as a view into my design process.)*

Starting Milestones

I. Concept sketches are drawn and preliminary electric components are selected.

II. Preliminary mechanical designs are chosen, and reasoning is provided. Basic circuit diagram is drawn.

III. Preliminary design is modeled in Fusion 360. Electronic components are ordered, and the structure pieces are 3D printed.

IV. Basic code for movement is written.

V. Initial prototype is assembled. Code is debugged and design changes should be noted for second iteration.



Image: Enemies | TUNIC Game Wiki

Preliminary Electric Components

Raspberry Pi

I prefer to use a Raspberry Pi for this project as opposed to an Arduino. I want to practice my mechatronics and coding skills, and I believe that this will work well with my design. Additionally, the Raspberry Pi has more capabilities that may be expanded upon if I want to add any additional goals for the spider robot.

PCA9685 Board

Provides PWM signals to the servo motors. Allows the Raspberry Pi to control multiple servos at once and move the robot.

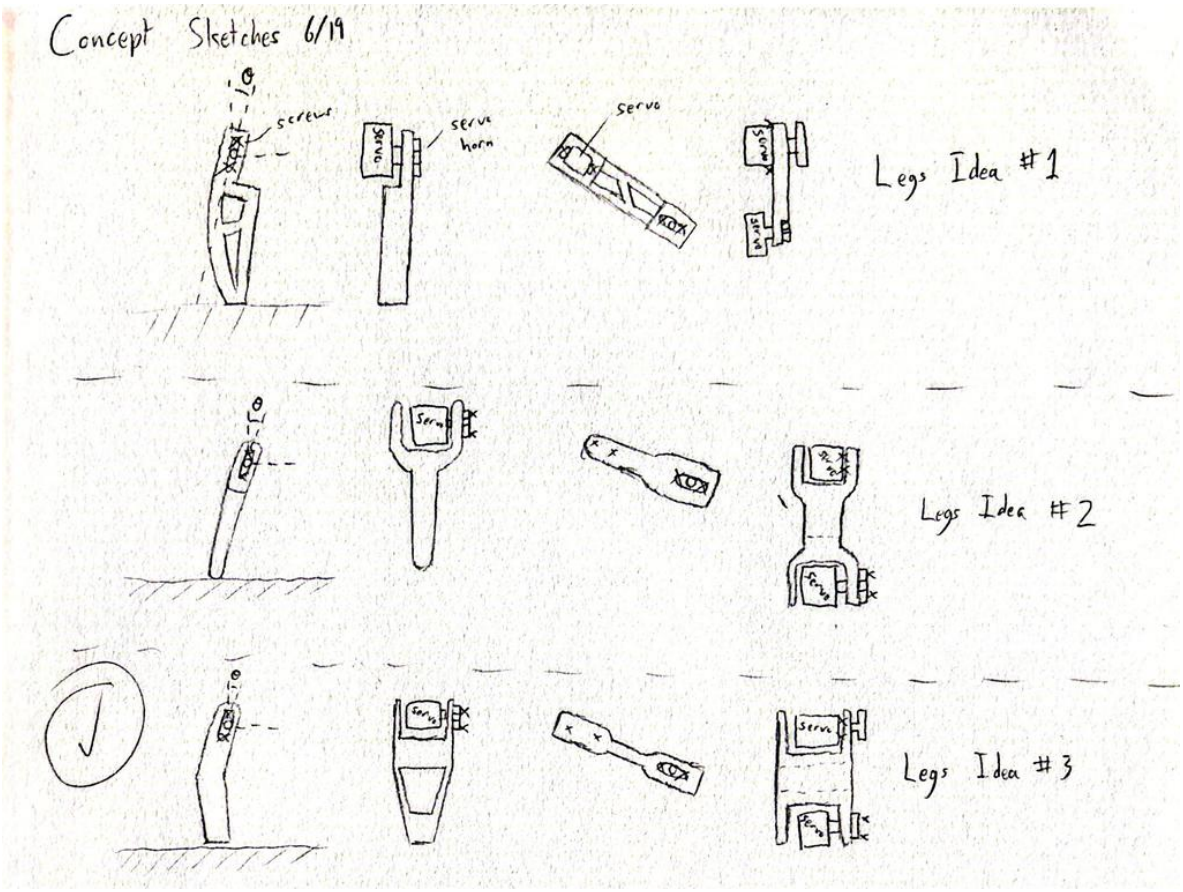
TowerPro SG92R Servo Motors

Smaller servo motors that will support the movement of weaker joints. There will be a total of eight on the robot, with two servos moving each leg.

TowerPro SG-5010 Servo Motors

Larger servo motors that are used in the joints responsible for holding up the weight of the whole robot. There will be one in each leg, closest to the base of the spider.

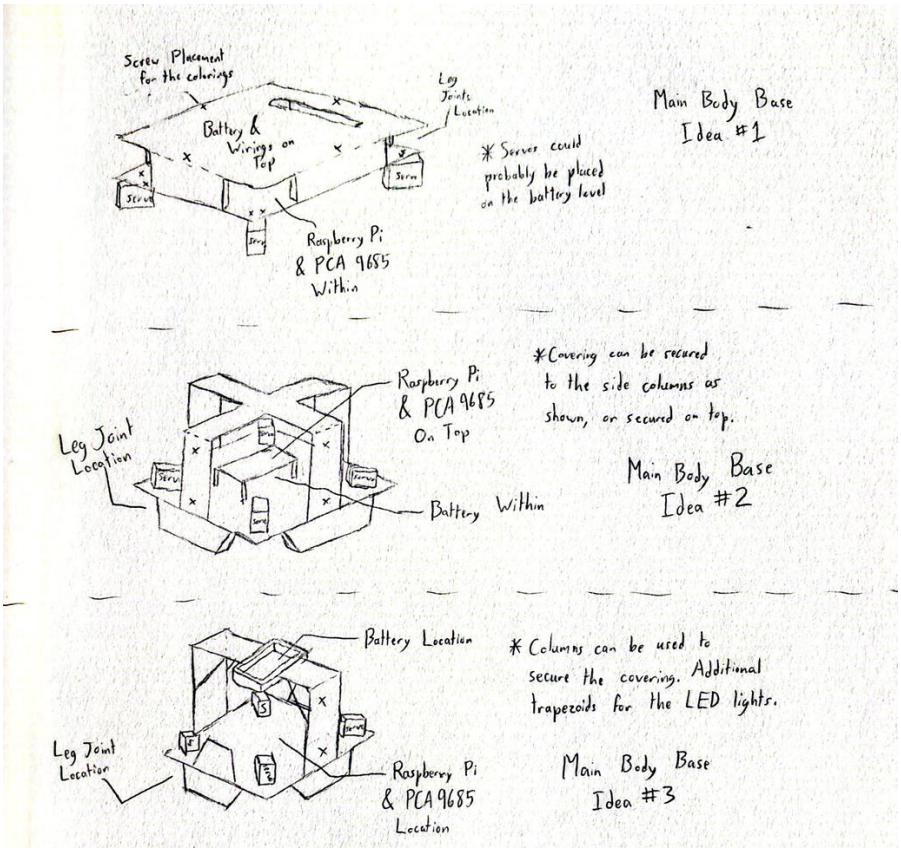
Concept Sketches



Design Considerations for Legs:

1. The angle at which the legs rest on the ground determines the moment that the servos will need to compensate for. This moment, caused by both the angle and overall weight of the robot, will be at its greatest when only two legs are in contact with the ground.
2. Contact area with the floor will increase the robot's balance.
3. Leg parts should not break or be overly strained during movement or rest.
4. Design should allow servo motors to be securely fastened while remaining free to move joints.

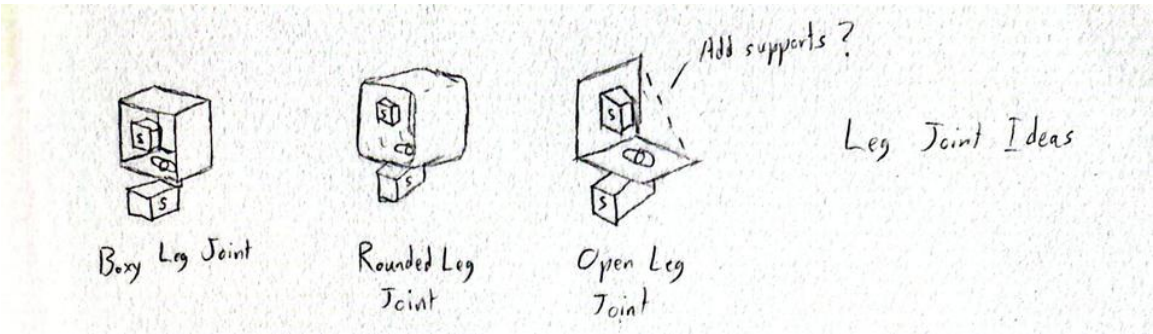
I like idea #3 the most. It has a flat area at the foot, which helps with balance, as well as being structurally ideal in servo integration and material integrity. The design has a higher adaptability for reiteration: supports and thickness can be added if the legs begin to buckle, and the angle of the front component can be changed to prevent unnecessary torque on its servo.



Design Considerations for Base:

1. Should keep design light and efficient to reduce straining leg servo motors.
2. Should allow easy organization of electric components and wirings.
3. Must allow for the covering to become securely fastened.
4. Location of components should be symmetrical in weight. This allows the robot to maintain balance more easily.

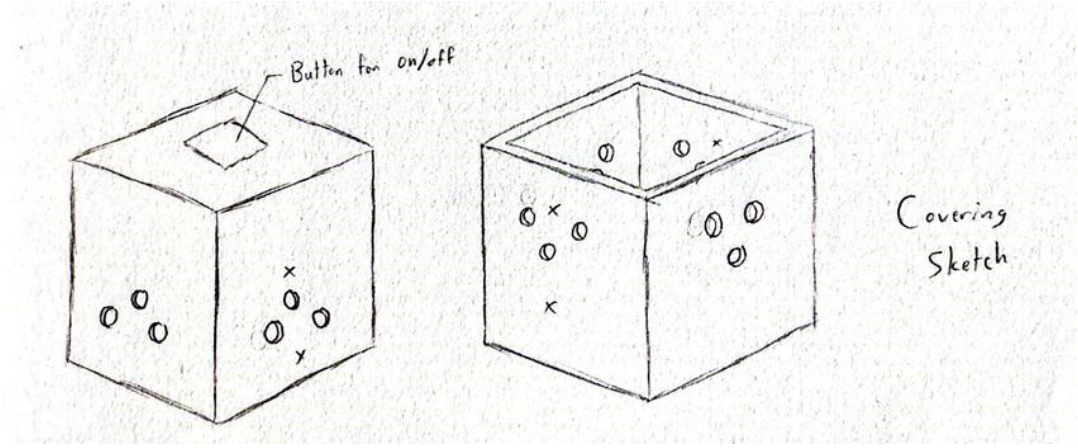
Idea #3 draws me in the most, because it fulfills each of my design considerations while remaining open enough for repairs and modifications. It keeps servos in the interior of the robot, while offering places for the LEDs to be placed. I may use idea #2's small table component, but that depends on if I need to use a breadboard for certain components.



Design Considerations for Leg Joints:

1. Leg joints will be the most load-bearing component of the robot. Must be sturdy and capable of withstanding the robot's weight.
2. Should be spacious enough within to house its respective servo motor.
3. Must integrate the base and leg together smoothly.

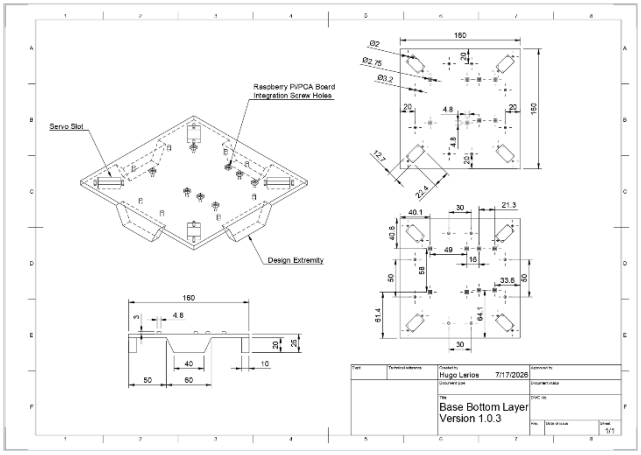
Most likely, I will choose the boxy or rounded designs for their simplicity and symmetry. If the joint is at risk of fracture, additional supports can be created to disperse loads caused by the servos' movements.

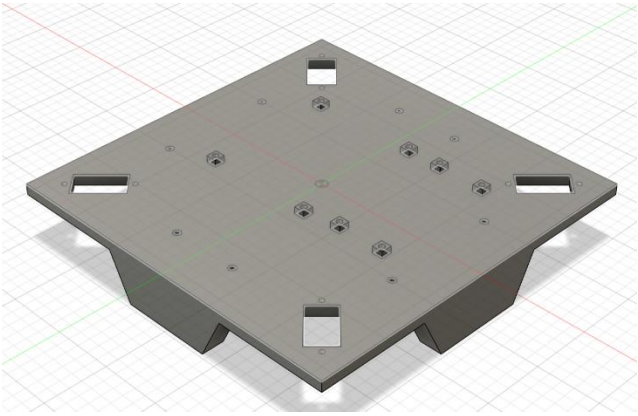


Given that the covering's only purpose is to be a decorative piece that resembles the inspiration of the project, it's best if the design remains simplistic. It should be a removable component for providing maintenance to the robot, and features a spot for an on/off button to be placed if desired.

Of note, however, if I plan to add a button for the power, thought must be given to how that aspect of the robot's internal circuitry will be wired.

Preliminary Design Component Models

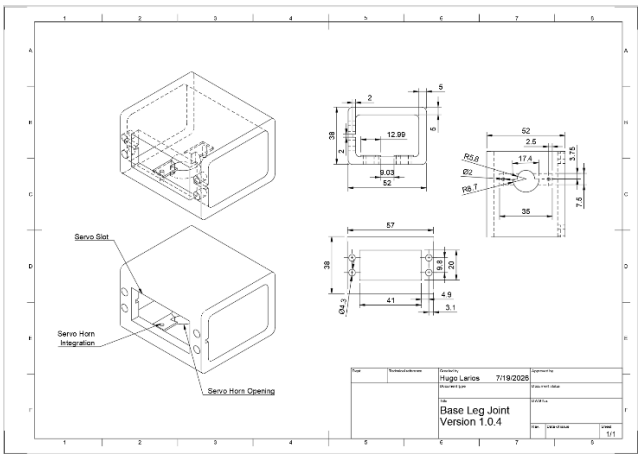


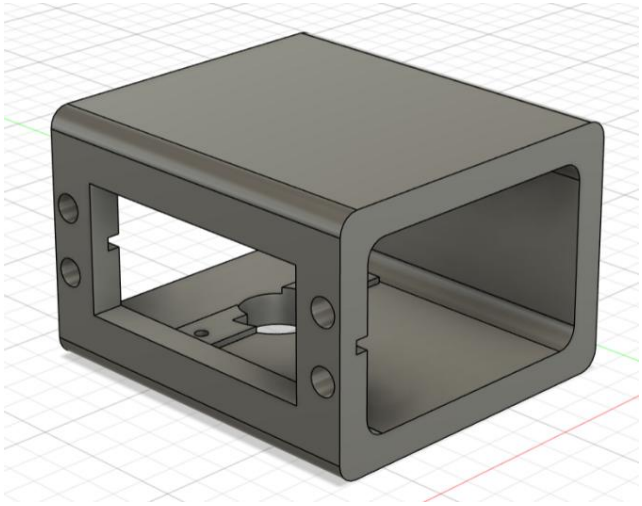


Base Platform Preliminary Design Choices:

- Servo slots were fitted into each of the four corners of the Base Platform. Servos would be oriented such that their wires were facing internally and their servo horn operating on the bottom side of the Base Platform.
- The Raspberry Pi and PCA board were noted to have small solder bits, preventing either from remaining flush with a flat surface. Therefore, small protrusions with screw holes were added, so that they may be secured to the Base Platform.
- Screw holes for the Base Platform Idea #3 breadboard support were created to allow for proper integration and security.
- Underside protrusions were added for purely cosmetic reasons, to maintain the spirit of the game enemy that inspired this robot's overall design.

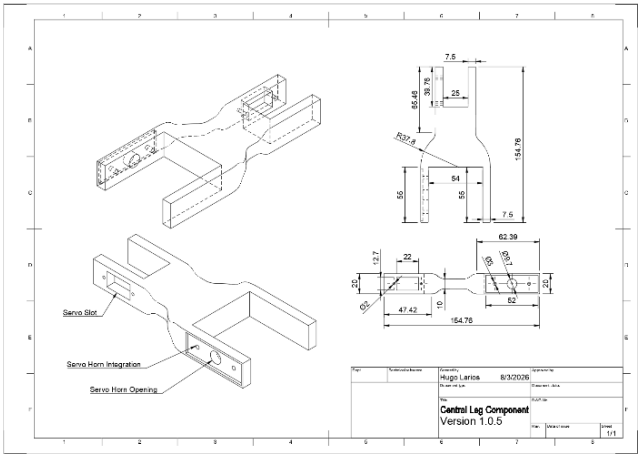
- The overall dimensions of the Base Platform were selected to allot appropriate space to house internal electrical components without overcompensating. Keeping smaller overall dimensions helps remove extraneous weight considerations and keeps components within creatable dimensions for most 3D printers.

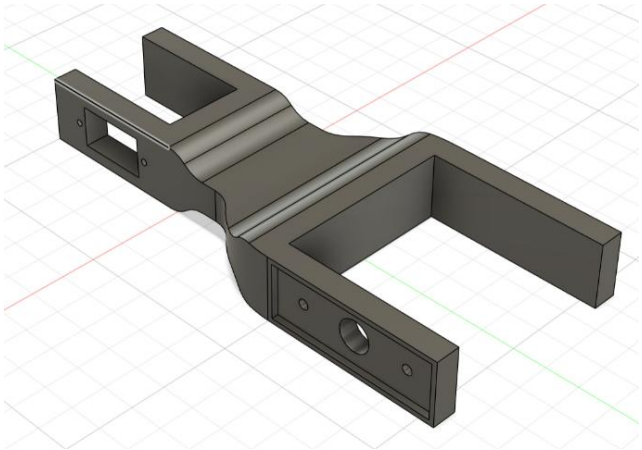




Base Leg Joint Design Choices:

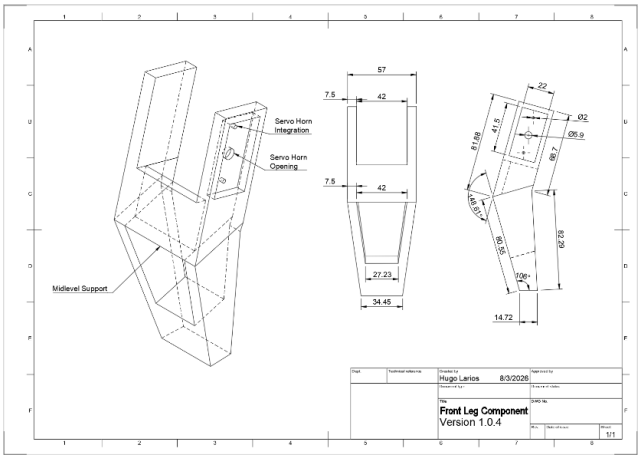
- Servo slot was placed on the side for proper axis of rotation for the Central Leg Component. Small divots were placed on the interior of the servo slot to fit better with the selected servo's geometry.
- The hole for connection to the Base Platform servos is made up of two differently sized half circles. The smaller half circle is meant to hug the central hub portion of the servo's axis of rotation, while the larger half circle is meant to allow this central hub portion to freely rotate to the necessary angles.
- The connection point between the Base Leg Joint and Base Platform is slightly raised to better affix the required servo horn to the component.
- Overall dimensions are selected to snugly fit the larger servo within. Kept no larger than necessary, to avoid unnecessary weight in the design.

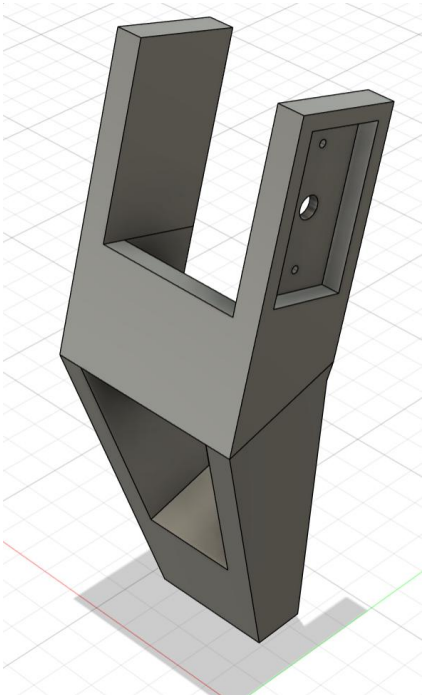




Central Leg Component Design Choices:

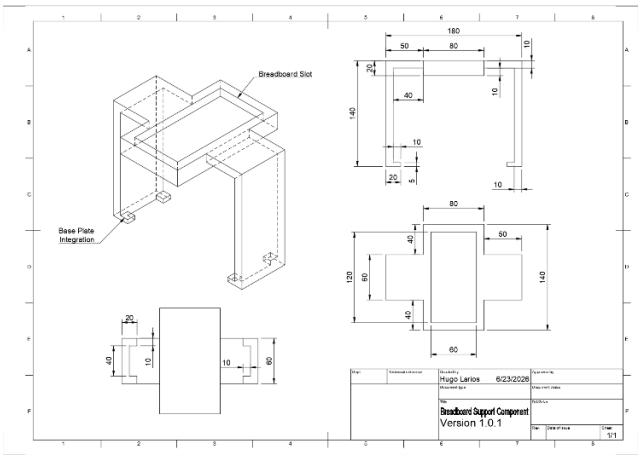
- Widths of the Central Leg Component's ends were determined by the size of the small servo for the servo slot, and size of the Base Leg Joint.
- Connection point thickness between Central Leg Component and Base Leg Joint is set to account for integration geometry. Screw holes are in place to connect with servo horn.
- Thickness between the two ends is decreased slightly, to save on additional weight while maintaining structural integrity.

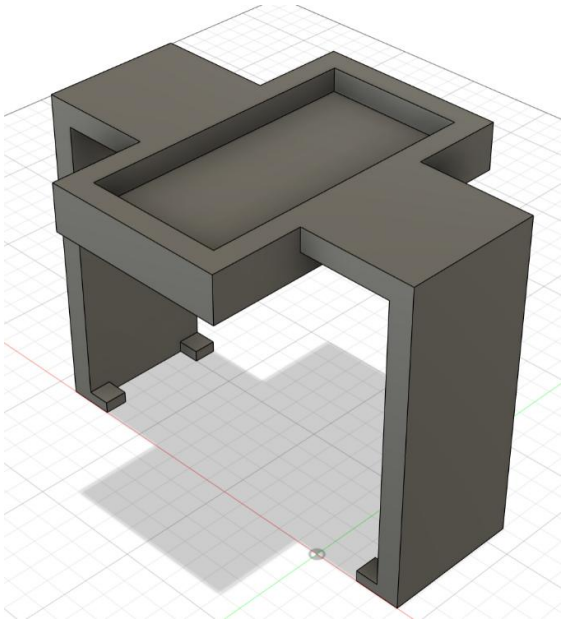




Front Leg Component Design Choices:

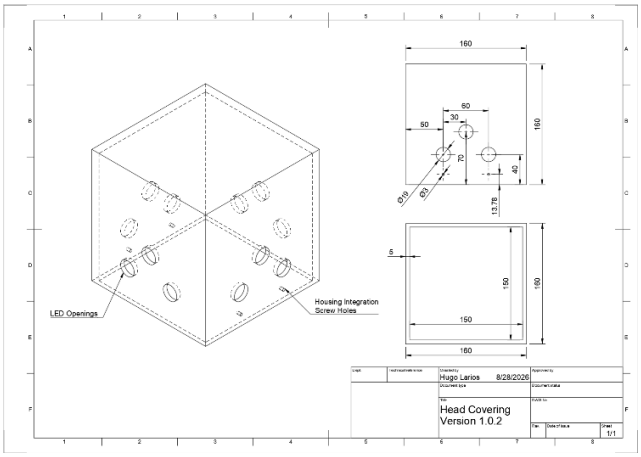
- Dimensional width is designed to account for Central Leg geometry with some amount of tolerance.
- Connection point thickness between Front Leg and Central Leg is adjusted to properly secure with servo horn without interfering with Central Leg geometry.
- Connection point is aligned such that it sits above the bottom end of the Front Leg. This is done to reduce the natural moment around the joint as the robot rests.
- Central connection between the left and right side of the Front Leg is added to increase structural integrity of the leg.

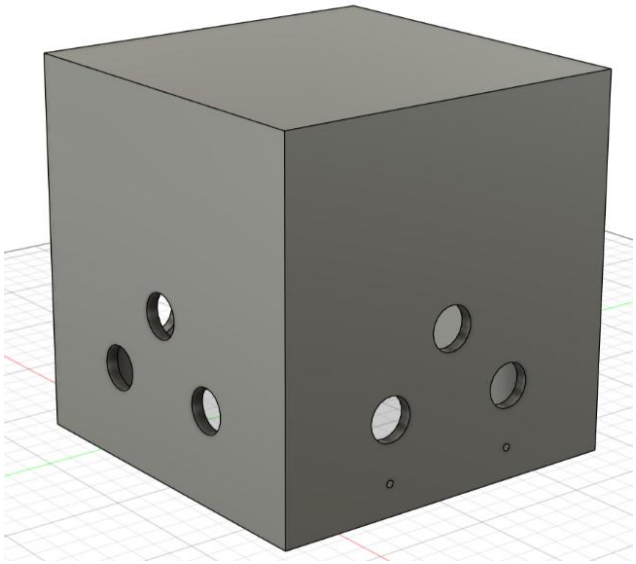




Breadboard Support Component Design Choices:

- Dimensions of the top portion are set to house the average size of a breadboard. The height of the overall Breadboard Support is set to slightly below the roof of the Head Covering.
- Two arms reach down with connection points that attach securely to the Base Platform.
- Thickness is set to an arbitrary amount and will be refined to reduce the overall weight of the component.





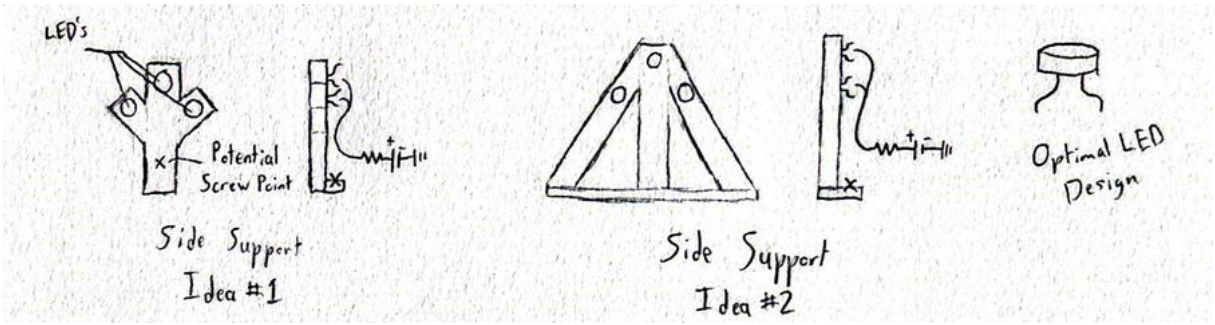
Head Covering Design Choices:

- The Head Covering has three holes on each face for each of the three LED “eyes” that will appear on each face.
- Screw holes are added to two of the main faces to allow integration with the Base Platform and its respective side supports.
- The LED “eyes” are placed symmetrically and spaced comfortably around each other. Specific spacing dimensions were arbitrary and were chosen because they look aesthetically pleasing.

Design Updates

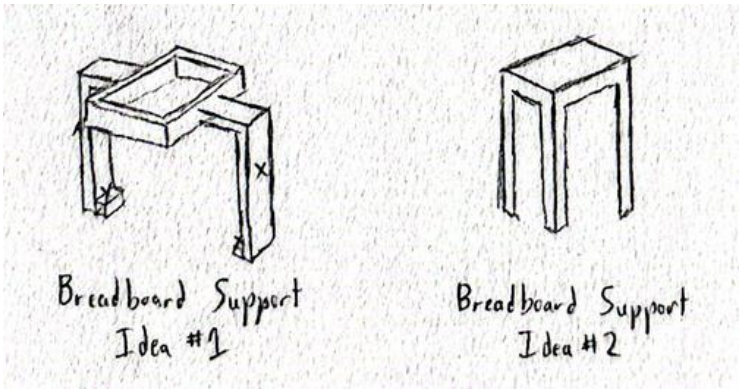
I've finished the CADs for the legs, so now I need to work on the base components. While modeling, I've converted my conceptual base design into three separate parts, so that the electronic integration and 3D printing can be done more efficiently. So far, I've separated it into the bottom base (holding servos and Raspberry Pi), breadboard holder (for the button and installation of the covering), and side supports (to hold the LEDs).

I am adding this entry because I want to come up with alternative design options for the breadboard holder and side-supports. Both components should be designed with electronic integration considered to be priority, since neither should endure any significant load during operation.



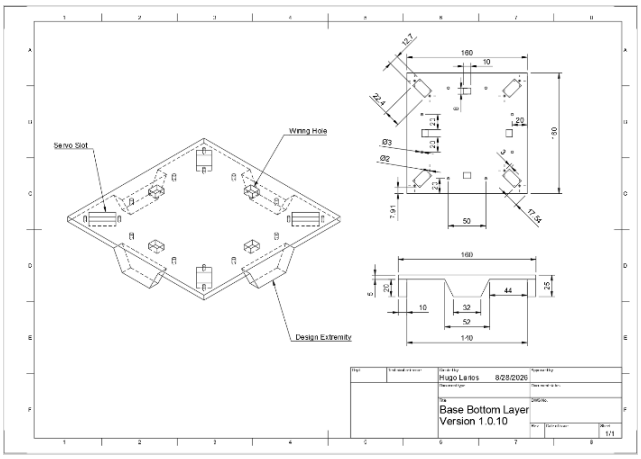
Design Considerations for Side Supports:

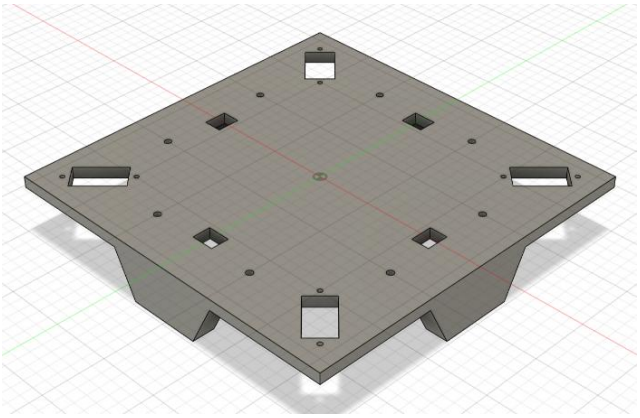
1. Space must be available for LEDs to be placed. Wirings must then follow along the interior-oriented face.
2. Must be easily integrated as its own component to the bottom base component.
3. May function as the cover's main securing component. This may free the breadboard holder to only hold the breadboard instead of acting as additional holding support for the cover.



Additional Note: Though the above sketches show my initial ideas for the Breadboard Support, I found that it was simpler to design a platform that simply rests atop the Side Support components. This generally reduces the need of an additional complex component, and can instead act as a simple plate that holds multiple electrical components. Through CADing the side supports on Fusion 360, I also found that my initial ideas of reducing mass by having the center portion branching out or using a triangular frame, was mostly unnecessary. The amount of mass it saves is relatively negligible compared to the dimensions of the LEDs and where they will be placed.

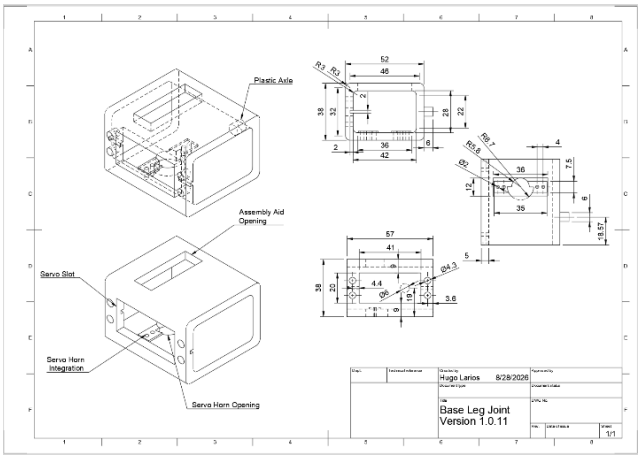
Prototype Design Components

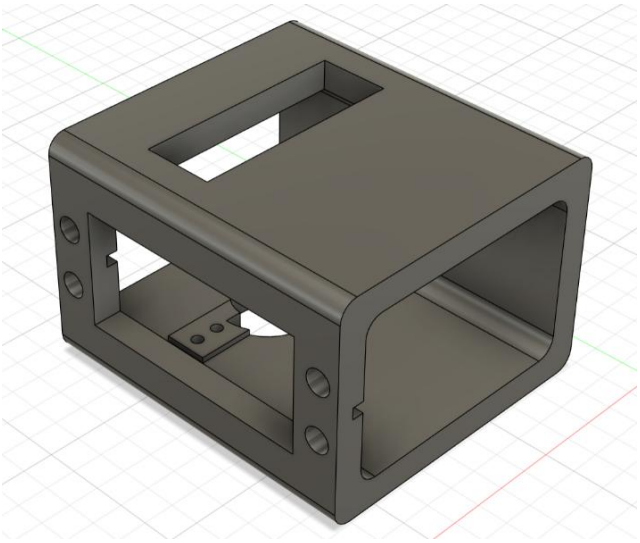




Base Platform Design Updates:

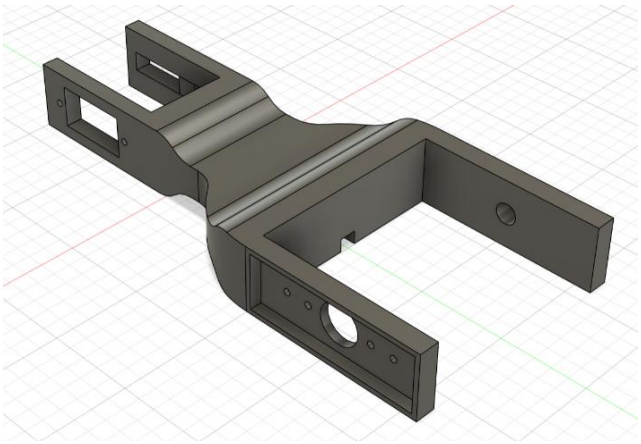
- Screw holes for small servo motors were readjusted. I found the original model for my servo motors was slightly inaccurate to real-world dimensions.
- Screw holes for LED holders are finalized and made to integrate with those components.
- Wiring holes were added around the base platform, to allow leg component servo motors to connect with interior Raspberry Pi.
- Bottom extremities were resized so as not to restrict the freedom of motion for the base leg joints.
- Protruding screw holes were removed, as Raspberry Pi now has black-cased housing, and the PCA board has been moved upwards on the new Central Platform component.

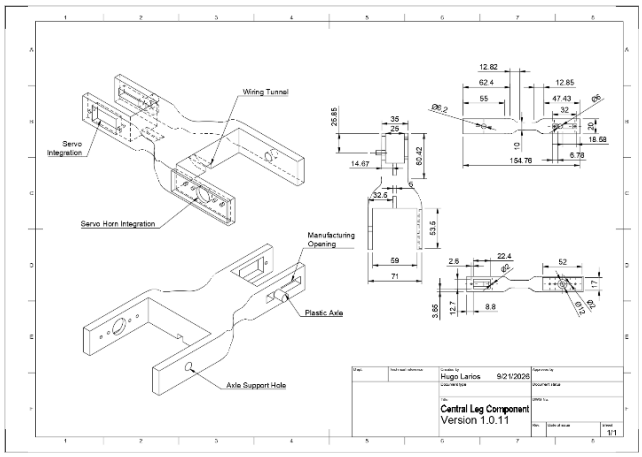




Base Leg Joint Design Updates:

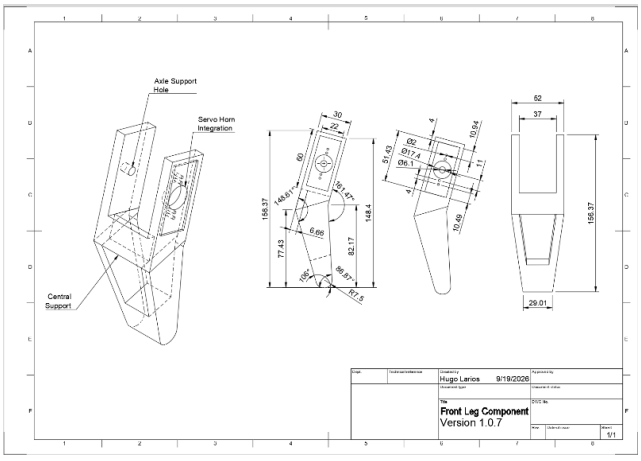
- Slightly resized to better accommodate placing the large servo motor within.
- Area above Base Platform integration is cut, to allow ease of assembly. Servo horn can now be screwed in more easily.
- Additional screw holes for Base Platform integration were added, to better secure servo horn to the component.
- Screw holes for servo integration were resized, due to a lack of proper tolerancing beforehand.

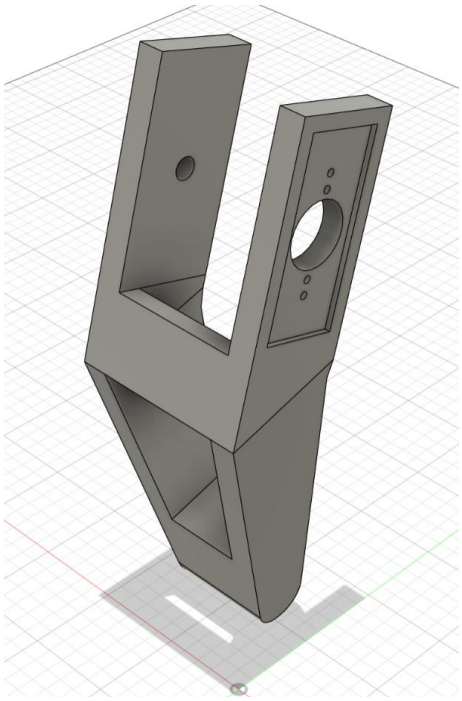




Central Leg Component Design Updates:

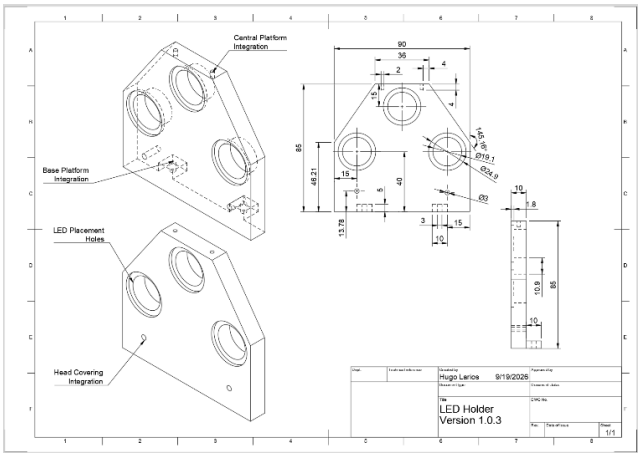
- Significantly resized to properly fit the base leg joint.
- Integration with servo motor from base leg joint is finalized, leaving two potential screw holes and an open circle to allow for proper rotation.
- Screw holes and servo housing were resized to better accommodate the servo integration.
- An additional slit was cut behind the servo integration point, to allow ease of assembly. Can now secure the servo with screws more easily.
- A small amount of material was cut at the bottom of the component for managing wires.

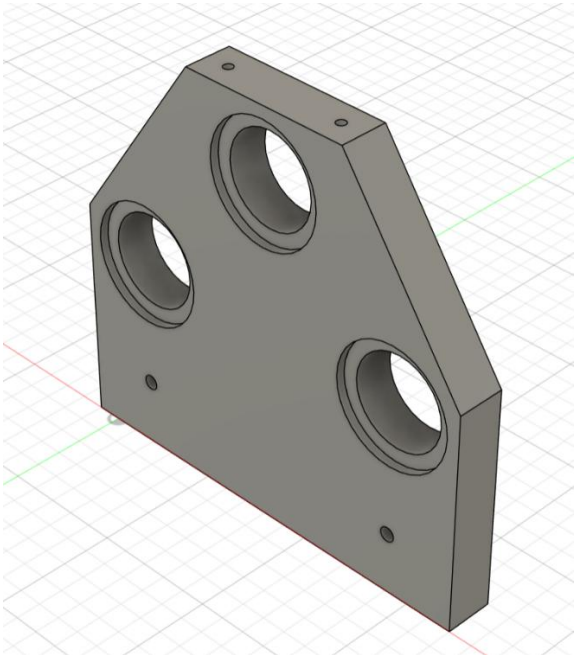




Front Leg Component Design Updates:

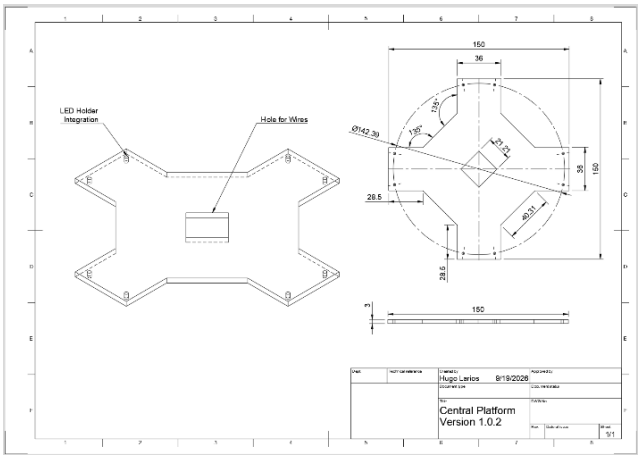
- Slight resizing of leg width, to compensate for changes made to the Central Leg Component.
- Size for servo integration hole was dramatically increased, to allow for full freedom of movement when the leg is extending or retracting.
- Servo integration holes were added, and the depth of the integration was made to be shallower. The servo integration between the Front Leg Component and the Central Leg Component is different than that of the Base Leg Joint and Base Platform, so resizing was needed to correct this.

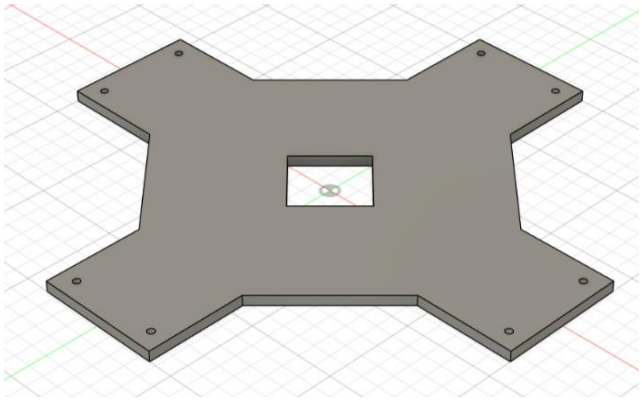




LED Holder Design Choices:

- Scrapped the original breadboard support idea in favor of the LED Holders. Each will house the 19mm LEDs and allow for connection between itself and the overall covering of the robot.
- Doesn't follow its preliminary drawings, but this is because it is a simpler shape that is more secure.
- Screw holes are added at the top to allow integration with the Central Platform.
- All four sides of the Base Platform will have a respective LED Holder.
- Behind the front-face of the component, there are screw holes that allow integration with the Base Platform.

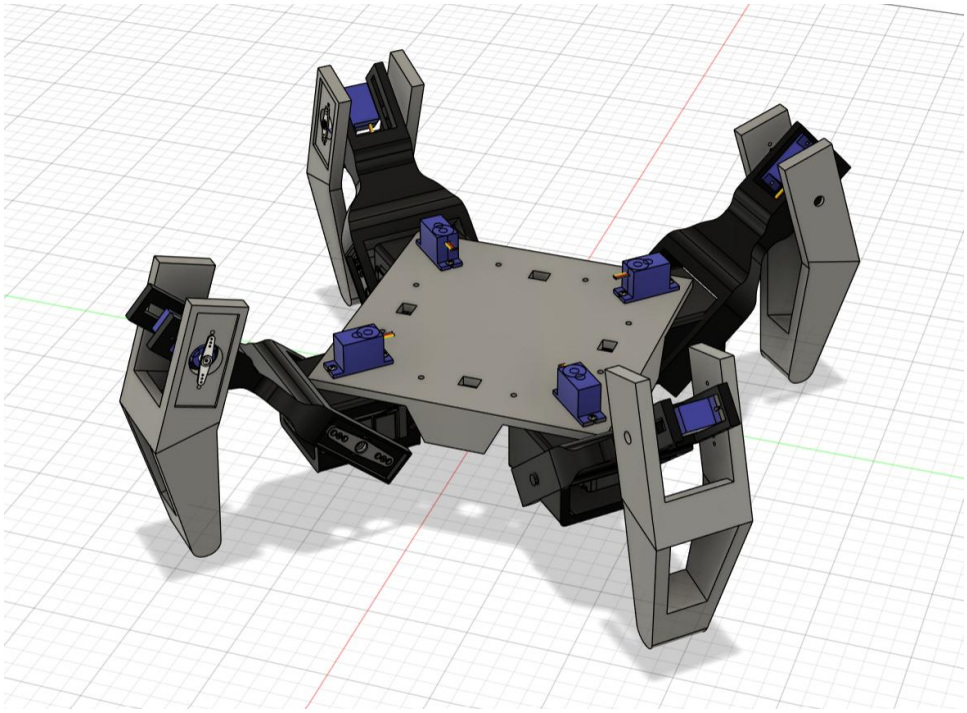




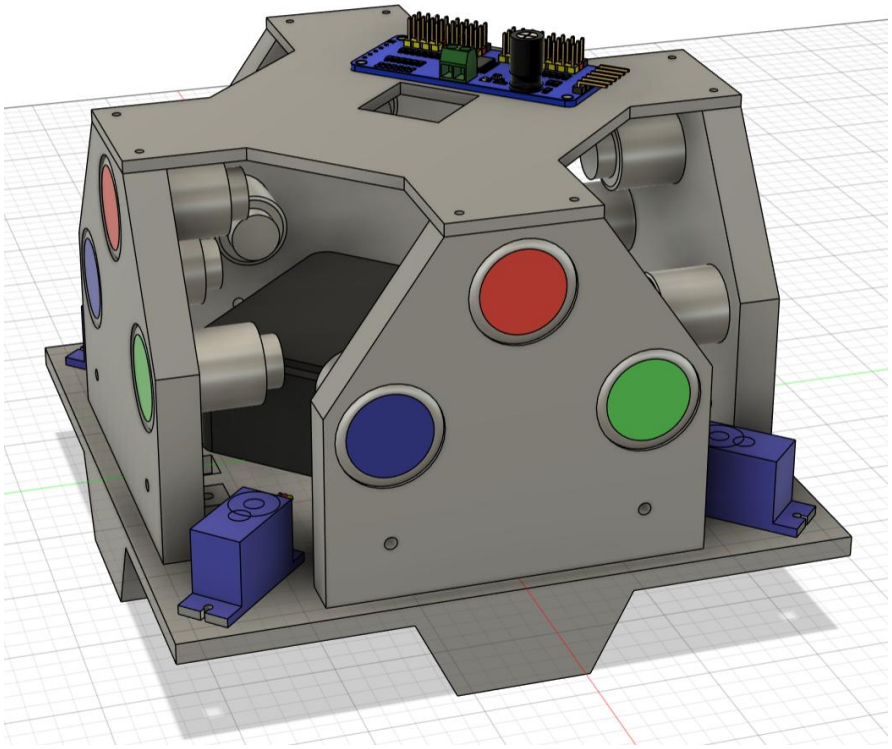
Central Platform Design Choices (Unfinished):

- Made to connect with all four LED Holders surrounding the Top Housing of the robot.
- Features a hole in the center to allow for wires to pass through to components resting atop the Central Platform.
- Still missing exact location and integration points with electronics. CADs for the batteries, voltage regulator, and thought for specific wire placements are currently missing. This section will be updated soon.

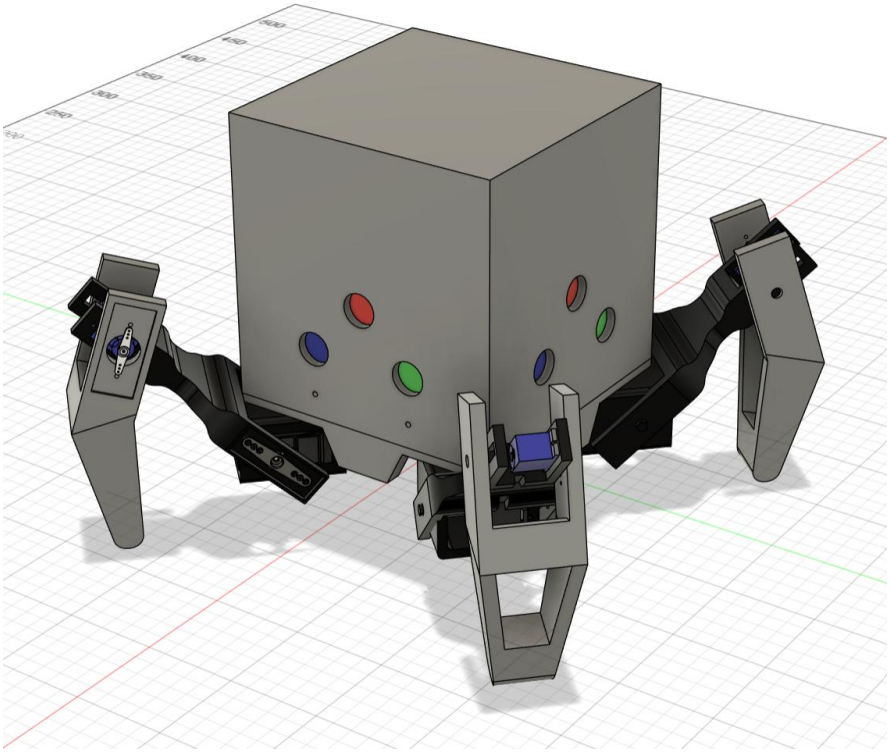
Prototype Design Assemblies



Bottom Leg Assembly



Top Housing Assembly (Current)



Complete Robot Assembly (W/ Covering)

Circuit Design

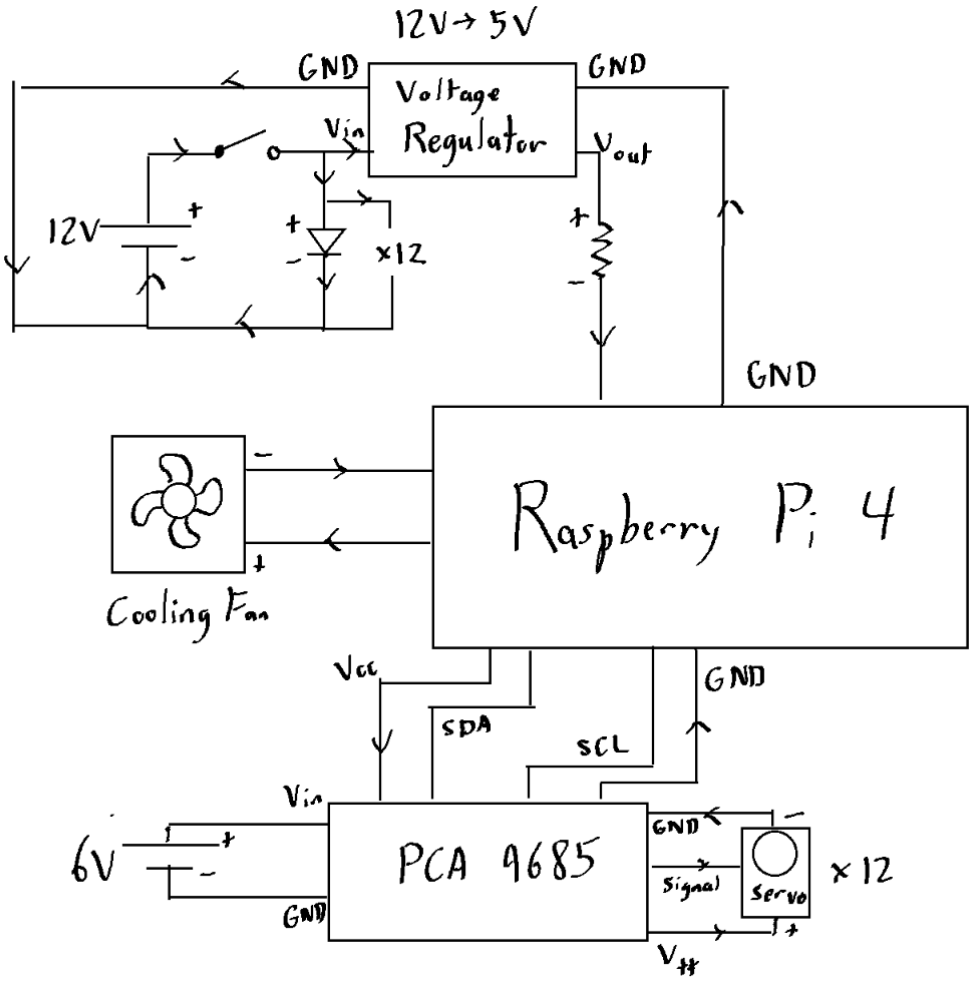
Prototype Components:

19mm LEDs – rated for 12V–220V, ≤ 20mA, 50–60Hz

- TowerPro SG92R – rated for 4.8V (3V to 6V DC), 10mA minimum, 100mA to 300mA, stall of 650mA
- TowerPro SG-5010 – rated for 5V (4.8V to 6V DC), 10mA minimum, up to 1.2A at 6V

Thought Process:

The total circuitry of the robot may need to be divided between two power sources, due to the servo motors' tendency to fluctuate current draws. My idea is to use a ≥12V battery for the LEDs and Raspberry Pi, while I use a 6V battery pack for the servo motors. The Raspberry Pi is only rated for ~5V, so I'll use a voltage regulator and resistors to step down the battery voltage. LEDs shouldn't need resistors, as they appear to have internal limiters.



Circuit Diagram

Manufacturing Plan

Because the bottom leg assembly has more movement/dynamics complexity, the best course of action would be to manufacture this half first and test. Once the robot's bottom half is created and a general program script for movement and balance is created, then the top half will be manufactured. All manufactured components were 3D printed using Bambu Lab printing software and made of basic PLA at 10-15% infill density.

Bottom Leg Assembly

Bottom leg assembly involves the following components:

- 4 Front Leg Components
- 4 Central Leg Components
- 4 Base Leg Joints
- Base Platform
- 8 TowerPro SG92R servo motors
  - 8 respective servo horns
  - 16 M2 screws of 7mm length
  - 8 respective servo horn screws
- 4 TowerPro SG-5010 servo motors

- 4 respective servo horns
- 8 M2 screws of 7mm length
- 8 M4 screws of 13mm length
- 8 M4 screw nuts
- 4 respective servo horn screws

For the case of my prototype and its fasteners, I needed two Phillips-head screwdrivers and a hex (Allen) wrench. Total set-up and components are shown below.



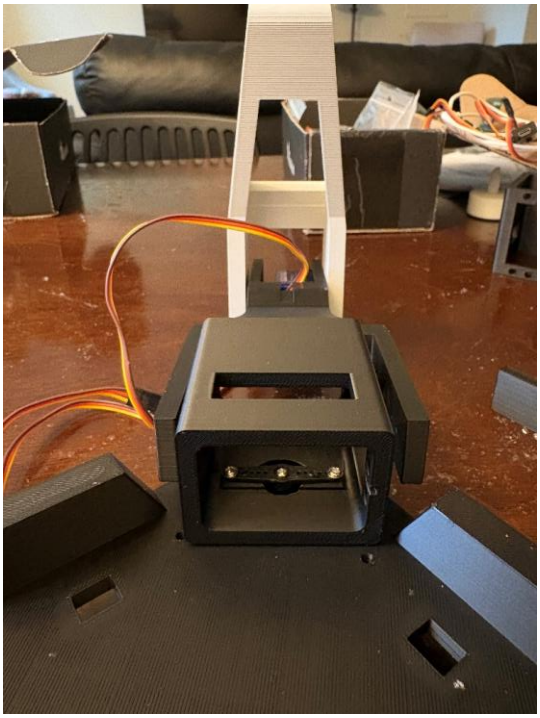
Components and tools for the Bottom Leg Assembly

The steps for manufacturing are as follows:

1. Join the Central and Front Leg components together. Install the TowerPro SG92Rs into their respective places between both components.
2. Once servo motors are installed, secure the components together using the servo horn and respective screws. Align the servo horn so that the leg has ample freedom of motion around its joint. Attach the Base Leg Joint at the bottom of the Central Leg component and set down. Repeat process for all legs.



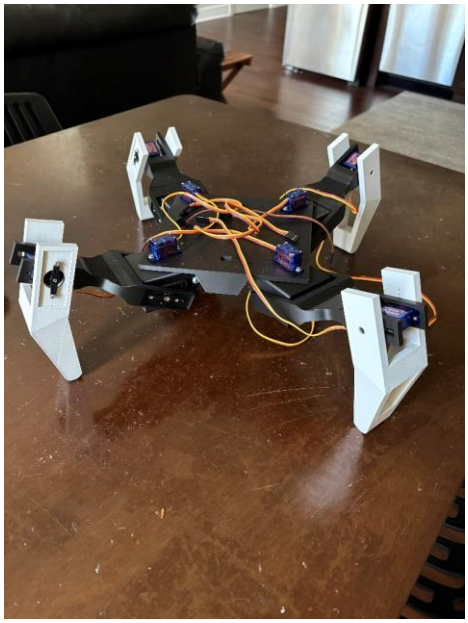
3. Install four SG92R servo motors to the appropriate side of the Base Platform, then set the platform down so that it rests on the servo motors. Take a leg and attach the appropriate hole onto a servo motor.
4. Using a servo horn, ensure that the servo motor is capable of the appropriate degrees of motion. Then, secure the leg to the base with said servo horn, and rotate to confirm solid attachment.



5. Insert SG-5010 servo motor within the Base Leg Joint cavity, positioned so that the wires are facing outward of the robot. Secure to the Base Leg Joint.
6. Attach servo horn onto the servo motor, then use it to secure the Base Leg Joint to Central Leg component using appropriate fasteners. Rotate to verify the desired freedom of motion.



7. Once leg is verified to be completely secured, funnel wires from both leg servo motors towards the base component. Use the wire openings on the sides of the Base Platform to allow for integration with the Top Housing Assembly.
8. Repeat Steps 4–7 for each leg, until the Bottom Leg Assembly is complete.



Top Housing Assembly

For the moment, the Top Housing Assembly has yet to be finalized. I'm still determining an adequate power supply for the PCA board and servo motors without risking battery sag from current requirements. Additionally, the Central Platform component still needs to be redesigned to accommodate for integration with electronic components and wirings.

Initial Code Testing

To check if the PCA board can adequately control the motion of the legs, a quick Python script was created to test the movement of a singular leg. Because only three servo motors needed to be controlled, four AA batteries were deemed adequate to power the PCA board and motors. The Raspberry Pi was powered by and controlled using my personal laptop. Test code for the leg is as follows:

```
# Import appropriate libraries
from adafruit_servokit import ServoKit
from time import sleep

# 16 channels on a standard PCA9685
kit = ServoKit(channels=16)

# Move each respective servo
kit.servo[0].angle = 45   # Move Base Leg Joint to middle position. Range of 5-85 degrees
sleep(1)
kit.servo[1].angle = 100  # Move Central Leg to middle position. Range of 45-130 degrees
sleep(1)
kit.servo[2].angle = 35   # Move Front Leg to Middle position. Range of 20-150 degrees
sleep(1)

# Move servos on channels 0, 1, 2 together
angles = {0: 90, 1: 45, 2: 180}

for channel, angle in angles.items():
    kit.servo[channel].angle = angle
    sleep(1)  # give them all time to physically reach position
```

Further steps and documentation will be added as the project progresses.